Product Aesthetic Design: A Machine Learning Augmentation

Replication and Extension Based on Burnap, Hauser, and Timoshenko (2023)

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March 9, 2026

Abstract

This report presents a detailed review and replication of the seminal work by Burnap et al. (2023) on quantifying product aesthetics using deep generative models. We first review the theoretical gap addressed by the authors—specifically, the challenge of measuring "Gestalt" aesthetic value versus discrete attributes. We then detail our replication attempt of their VAE-GAN architecture, highlighting critical instability issues such as gradient explosions and mode collapse ("The White Screen Phenomenon"). Finally, we propose a novel extension: shifting from training custom VAE-GANs to utilizing pre-trained Foundation Model Agents. We demonstrate a workflow integrating Stable Diffusion (Design), CLIP (Market evaluation), and Computer Vision (Operations constraints) to automate the discovery of Pareto-efficient designs.

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1 Introduction & Motivation

1.1 The Theoretical Gap: Gestalt vs. Attributes

A large fraction of empirical marketing models assume that consumers decompose products into a vector of attributes and then apply an additive (or separable) valuation rule, as in conjoint analysis and discrete-choice utility modeling. In contrast, *aesthetic appeal* is often *Gestalt*: an emergent property of a high-dimensional visual stimulus that is not well-approximated by a short list of hand-engineered attributes such as "headlight shape" or "grille size." Formally, if a design is represented by an image $X \in \mathbb{R}^{H\times W\times 3}$, then aesthetic valuation is a functional

$$v:\ \mathbb{R}^{H\times W\times 3} \to \mathbb{R},$$

which is nonlinear and interaction-driven (i.e., depends on complex compositions of parts). Standard attribute models effectively assume

$$v(X) \approx \sum_{j=1}^{p} \alpha_j \phi_j(X),$$

for a small set of pre-specified feature extractors $\{\phi_j\}$; the paper's premise is that for aesthetics, the relevant ϕ_j are high-dimensional and hard to hand-design, and the separability assumption is too restrictive.

A second challenge is *statistical*: aesthetic ratings are expensive and thus "thin." A convenient measurement model is

$$y_{i,r} = v(X_i) + \varepsilon_{i,r}, \qquad r = 1, \ldots, n_i,$$

where $y_{i,r}$ is respondent r's rating of design i, and firms often only observe the aggregate $\bar{y}_i = \frac{1}{n_i}\sum_r y_{i,r}$. With only a few hundred distinct designs, estimating a flexible $v(\cdot)$ is underidentified unless one can exploit additional structure and data.

This creates a practical "measurement problem" for management: design drives sales, but firms lack a scalable quantitative instrument (a "ruler") to evaluate and iterate designs early in the pipeline. Off-the-shelf computer vision systems trained on ImageNet-style tasks primarily learn *discriminative* representations for classification ("what object is this?") rather than *preference* or *valuation* ("how appealing is this design?"). Even if a pretrained CNN yields generic features, the mapping from those features to subjective semantic scales is not guaranteed to be learnable with only ~ 200 labeled designs. Burnap et al. (2023) address this by learning a probabilistic design embedding from large-scale unlabeled imagery and then learning the valuation map from scarce labels.

1.2 Problem Statement and Notation (as in Burnap et al. 2023)

Burnap et al. (2023) consider a semi-supervised setting with two datasets:

- **Rated set $\mathcal{R}$**: tuples $\{(X_i, m_i, a_i, y_i)\}_{i\in\mathcal{R}}$ where $y_i \in \mathbb{R}$ is an average consumer rating (e.g., aesthetic appeal on a Likert-type scale), a_i are observed attributes (e.g., brand, model year, viewpoint), and m_i is a segmentation mask that isolates the design from background.
- **Unrated set $\mathcal{U}$**: tuples $\{(X_i, m_i, a_i)\}_{i\in\mathcal{U}}$ without y_i, but typically with far larger $|\mathcal{U}|$.

In their automotive proof-of-concept, $|\mathcal{R}| = 203$ SUVs/CUVs (with thousands of consumer ratings aggregated to a vehicle-level mean), whereas $|\mathcal{U}| \approx 180{,}000$ high-quality images.

The objective is to learn a representation and two task-specific maps:

1. **Prediction (valuation):** a predictor $\hat{y} = f_P(h)$ that maps a learned embedding $h \in \mathbb{R}^K$ of the design to a rating, enabling high-throughput screening of concepts.

2. **Generation (design synthesis):** a generator/decoder $\hat{X} = f_G(h)$ that maps embeddings to images, enabling exploration in a smooth latent design space.

Crucially, the embedding must be *identifiable* from $\mathcal{R} \cup \mathcal{U}$ and *useful* for predicting y using only $\mathcal{R}$. This is a representation-learning analogue of the "blessing of unlabeled data".

1.3 Why a Probabilistic Embedding? (Regularization, Sampling, and Identifiability)

A deterministic embedding $h = f_E(X)$ is vulnerable to memorization when labels are thin: the encoder can map each training image to a distinct region in latent space, letting the decoder reconstruct perfectly while destroying the geometry needed for sampling or generalization. Burnap et al. instead learn an *embedding distribution*

$$q_{\text{enc}}(h \mid X, a) = \mathcal{N}\big(\mu(X, a), \operatorname{diag}(\sigma^2(X, a))\big),$$

and penalize deviation from a standard normal prior via a KL divergence. The probabilistic embedding serves three technical roles:

- **Capacity control:** KL-to-prior discourages arbitrarily complex codes and reduces overfitting.
- **Sampling semantics:** if real designs map near $\mathcal{N}(0, I)$, then sampling $h \sim \mathcal{N}(0, I)$ becomes meaningful for generation.
- **Smooth geometry:** the latent space becomes navigable; local movements correspond to coherent visual changes.

This is essential for the managerial use case: not only predicting which designs are appealing, but also generating plausible new candidates.

1.4 The Practical Crisis in Industrial Design (Why the ML Formulation Matters)

Industrial design pipelines are iterative and constraint-driven: early sketches are filtered by engineering feasibility (safety, aerodynamics), manufacturing constraints (tooling, assembly), and cost constraints. Without a quantitative measure of how these constraints impact consumer appeal, firms risk *aesthetic dilution*—incremental changes that erode perceived attractiveness while satisfying downstream constraints. Moreover, direct human evaluation is expensive: a single theme clinic can cost on the order of $100,000 and is typically deployed late, when major redesigns are costly.

Hence, the promise of Burnap et al. (2023) is a systems-level augmentation of the design funnel: (i) use ML to allocate scarce human evaluation budget toward high-potential designs, and (ii) use a controllable latent design space to accelerate iteration under constraints, with quantitative feedback available earlier and cheaper than traditional clinics.

2 Methodology: The AI Engine

2.1 Core Idea: Semi-supervised VAE with Adversarial Terms

Burnap et al. (2023) formalize three coupled models: a *predictive* model (ratings), a *generative* model (images), and an *encoding* model (probabilistic embeddings). Let the parameter blocks be β_P, β_G, and β_E, and $\beta = (\beta_P, \beta_G, \beta_E)$. The training objective is a sum of three loss components:

$$L(\beta) = L_{\text{pred}}(\beta_P) + L_{\text{gen}}(\beta_G) + L_{\text{enc}}(\beta_E). \tag{1}$$

The separability is motivated by a probabilistic factorization and approximate marginalization (details in the paper's appendix), but in practice each term is weighted for numerical stability and task balance.

2.2 Probabilistic Formulation and ELBO Intuition

A compact way to understand the method is via a latent variable model. Assume a prior $p(h) = \mathcal{N}(0, I)$ and a decoder likelihood $p_{\beta_G}(X, m \mid h)$ that reconstructs image pixels and masks. The VAE objective maximizes a lower bound (ELBO) on the marginal log-likelihood:

$$\log p(X, m) \geq \mathbb{E}_{q_{\beta_E}(h|X,a)}[\log p_{\beta_G}(X, m \mid h)] - D_{\mathrm{KL}}(q_{\beta_E}(h \mid X, a) \,\|\, p(h)).$$

The paper implements the reconstruction term with MAE, which corresponds to a Laplace likelihood model for pixel intensities and masks. The predictive model adds an additional likelihood term $p_{\beta_P}(y \mid h)$, also implemented with MAE (Laplace). This yields a unified view: the model learns *(i)* a latent design representation useful for reconstruction and *(ii)* a valuation mapping from latent space to semantic ratings.

2.3 Model Components (Operational View)

The architecture can be summarized as four interacting neural modules plus attribute imputation:

1. **Encoder E:** $(\mu_i, \sigma_i) = f_E(X_i, a_i; \beta_E)$ parameterizes $q_{\mathrm{enc}}(h_i \mid X_i, a_i)$.
2. **Generator/Decoder G:** $(\hat{X}_i, \hat{m}_i) = f_G(h_i; \beta_G)$ reconstructs a 4-channel output (RGB + mask).
3. **Predictor P:** $\hat{y}_i = f_P(h_i; \beta_P)$ predicts semantic ratings.
4. **Attribute head (within E):** produces $\hat{a}_{ic\ell}$ (probability of level ℓ for attribute c), enabling missing-attribute imputation and disentanglement.

Important technical distinction: unlike conventional GANs, the paper does **not** rely on a separate pixel-level discriminator. Instead, adversarial training is performed in the *embedding space* via KL-to-prior terms, with the encoder acting as the "critic".

2.4 Loss Terms (Paper-Consistent Technical Form)

(i) Predictive loss (rated data only). With $\hat{y}_i = f_P(h_i; \beta_P)$,

$$L_{\mathrm{pred}}(\beta_P) = \sum_{i \in \mathcal{R}} \left| y_i - \hat{y}_i \right|. \tag{2}$$

This MAE objective is robust and matches the paper's probabilistic interpretation as Laplace noise with unit scale.

(ii) Generative reconstruction loss (rated + unrated). Let D be the number of spatial pixels per channel. With a 4th "mask channel," the generator loss is

$$L_{\mathrm{gen}}^{\mathrm{recon}}(\beta_G) = \sum_{i \in \mathcal{R} \cup \mathcal{U}} \left(\frac{1}{3D} \sum_{d=1}^{3D} \left| x_{id} - \hat{x}_{id} \right| + \frac{1}{D} \sum_{d=1}^{D} \left| m_{id} - \hat{m}_{id} \right| \right). \tag{3}$$

The mask term is not cosmetic: it reduces spurious background shortcuts and forces capacity toward object geometry and styling cues.

(iii) Encoding loss: KL-to-prior + attribute cross-entropy (rated + unrated). The encoder is regularized by KL divergence and simultaneously trained to predict attributes:

$$L_{\text{enc}}(\beta_E) = \sum_{i \in \mathcal{R} \cup \mathcal{U}} \left(-D_{\text{KL}}(q_{\text{enc}}(h \mid X_i, a_i) \,\|\, \mathcal{N}(0, I)) + \sum_{c=1}^{C} \sum_{\ell=1}^{L_c} a_{ic\ell} \log \hat{a}_{ic\ell} \right) \tag{4}$$

$$= \sum_{i \in \mathcal{R} \cup \mathcal{U}} \left(-\sum_{k=1}^{K} \frac{1}{2} \Big(\mu_{ik}^2 + \sigma_{ik}^2 - \log \sigma_{ik} \Big) + \sum_{c=1}^{C} \sum_{\ell=1}^{L_c} a_{ic\ell} \log \hat{a}_{ic\ell} \right), \tag{5}$$

where $a_{ic\ell}$ is a one-hot encoding for attribute levels. The attribute term also provides a mechanism for *imputing missing attributes* via the predicted multinomial distribution.

2.5 Adversarial Training in Embedding Space (Min–Max Game)

To encourage that *generated* images encode like *real* images (and thus be sampleable from the prior), Burnap et al. introduce an adversarial KL term defined on generated images only. Sample $h_g \sim \mathcal{N}(0, I)$ (optionally conditioned on a_g), generate $X_g = f_G(h_g)$, and re-encode to $q_{\text{enc}}(h \mid X_g, a_g)$. Define

$$L_{\text{adv}}(\beta_E) = \sum_{g \in \mathcal{G}} D_{\text{KL}}(q_{\text{enc}}(h \mid X_g, a_g) \,\|\, \mathcal{N}(0, I)) = \sum_{g \in \mathcal{G}} \sum_{k=1}^{K} \frac{1}{2} \Big(\mu_{gk}^2 + \sigma_{gk}^2 - \log \sigma_{gk} \Big). \tag{6}$$

The generator is trained to *minimize* the KL-to-prior for generated images (so generated images map near the prior), while the encoder is trained with the *opposite sign* (so it attempts to separate generated from real in embedding space). This yields a two-player min–max equilibrium:

$$\min_{\beta_G, \beta_P} \Big(L_{\text{pred}} + L_{\text{gen}}^{\text{recon}} + \lambda_{\text{adv}} L_{\text{adv}} \Big), \qquad \min_{\beta_E} \Big(L_{\text{enc}} - \lambda_{\text{adv}} L_{\text{adv}} \Big),$$

trained iteratively. Compared to pixel-discriminator GANs, this preserves an explicit probabilistic model and directly targets a smooth, controllable latent space—the managerial object of interest.

2.6 Latent Geometry and Spherical Interpolation (slerp)

When exploring designs, the paper samples points in embedding space and *moves smoothly* between them. In high dimensions, Gaussian samples concentrate near a hypersphere, motivating spherical linear interpolation (slerp). For normalized vectors u, v with angle $\theta = \arccos(u^\top v)$,

$$\text{slerp}(t; u, v) = \frac{\sin((1-t)\theta)}{\sin \theta} u + \frac{\sin(t\theta)}{\sin \theta} v, \qquad t \in [0, 1].$$

This produces intermediate embeddings with comparable norm, which empirically improves visual continuity and avoids norm-shrinkage artifacts common under linear interpolation.

2.7 Practical Stabilization and Training Engineering (What Makes It Work)

The paper emphasizes that training is nontrivial and requires stabilization:

- **Spectral normalization:** used in each block to prevent unbounded backpropagated gradients and reduce catastrophic failures (e.g., white-pixel outputs).
- **Variance bounding and floating-point safeguards:** explicitly constraining variances in KL terms and mitigating numerical instability introduced by logarithms and norms.

- **Loss reweighting and annealing:** KL-related terms are scaled to be roughly 1/10–1/20 of reconstruction losses and annealed from 0 to their target magnitudes; a fixed margin controls the KL gap between observed and generated images.
- **Progressive training (ProGAN-style):** training begins at 4×4 resolution and progressively grows to 512×512, linearly blending resolutions during transitions; the paper reports $\sim 10^6$ iterations per stage.

Computationally, training can take 1–2 weeks on a multi-GPU workstation (e.g., 4× Tesla V100 or 4× Quadro RTX 8000), with batch sizes constrained by GPU memory (the paper reports that larger batch sizes improve stability at high resolutions).

3 Key Empirical Results (Burnap et al. 2023)

3.1 Predictive Ability: Evaluation Protocol and Benchmarks

The predictive task is to forecast mean aesthetic ratings for held-out vehicles using images only. To avoid leakage, the unit of evaluation is the *vehicle/model* rather than individual images. The paper reports mean absolute error (MAE) and compares against benchmarks of increasing sophistication, with hyperparameters tuned on the same training/validation splits to ensure fairness.

The benchmark ladder is conceptually informative:

- **Uniform baseline:** constant prediction using the median rating in the training set (a conservative baseline).
- **CV+RF baseline:** hand-engineered computer vision features (classical pipeline) fed into a random forest regressor.
- **Pretrained CNN:** VGG16 with fine-tuned final layers, representing a strong transfer-learning baseline.
- **Proposed model:** semi-supervised probabilistic embedding + rating prediction.

Results (Table 2 in the paper) show that the proposed model achieves MAE = 0.350 (s.e. 0.038), outperforming CV+RF (MAE = 0.446) and VGG16 fine-tune (MAE = 0.405), and improving by 43.5% relative to the constant baseline (MAE = 0.620). The technical takeaway is that semi-supervised embedding learning is decisive in the "$|\mathcal{R}|$ small, $|\mathcal{U}|$ large" regime.

3.2 Generative Capability: What the Paper Can (and Cannot) Prove

Burnap et al. explicitly acknowledge that definitive causal market validation of generated designs is infeasible (no A/B test to market launch). They triangulate value via four lenses: (i) face validity and controllable morphing, (ii) consumer evaluations on generated pairs, (iii) managerial judgment, and (iv) "foresight" (similarity to vehicles launched after the training window).

Face validity and controllability. They sample embeddings conditioned on attributes and traverse the space via slerp, producing realistic morph sequences (SUV/CUV side views) that mimic the iterative nature of design studios.

Consumer evaluations (targeted generation). They generate 50 targeted images (25 predicted high-appeal, 25 predicted low-appeal), then run paired-comparison tests with screened "intender" respondents. After pre-registered attention filtering (IMCs), the predictor and consumers agree **74.0%** of the time on which images are more appealing, suggesting that generation is not only realistic but also aligned with the intended semantic ordering.

3.3 Foresight: Extrapolating Design Trajectories

A qualitative yet strategically important result is design "foresight": trained on MY2010–2014 SUVs/CUVs, the generator produces designs exhibiting styling cues that become more prevalent by MY2020. Interpreted through the representation-learning lens, this suggests that aesthetic evolution is constrained to a low-dimensional manifold; moving along dominant latent directions produces plausible future-like variants, though the claim remains qualitative rather than a formal forecasting guarantee.

3.4 Generalization Beyond Automotive: Dining-Room Chairs

To demonstrate domain generality, the paper repeats predictive tests on dining-room chairs (CAD-like renderings with sparse labels). Table 3 reports MAE improvements:

- Baseline MAE = 0.480 (s.d. 0.011),
- CV+RF MAE = 0.430,
- VGG16 fine-tune MAE = 0.434,
- Proposed model MAE = 0.423 (11.9% improvement).

The gains are smaller than in automotive, consistent with lower-fidelity imagery and potentially noisier semantic signal, but the result supports portability of the probabilistic-embedding template.

3.5 Recalibration to Alternative Semantic Scales

A final empirical point is that the framework is not tied to "appeal" alone. The same embedding-and-predictor pipeline can be recalibrated to other semantic metrics (e.g., innovativeness; traditional vs. modern). Technically, this emphasizes that the *embedding* is the reusable asset, while the predictor head f_P is metric-specific. In managerial terms, the model becomes a general-purpose "semantic meter" for multiple design objectives.

4 Replication Study: The "Classical" Approach

4.1 Replication Objective and Methodology

We attempted to replicate the Burnap et al. (2023) pipeline using a dataset of chair images to quantify aesthetic value in furniture design. The dataset is open sourced and is applied in the original paper as an example of the generalization of the proposed machine learning framework. The architecture employed was the standard VAE-GAN, aiming to compress chair images into a latent "DNA" (z) and reconstruct them. We used the codes and data downloaded from the link in the paper.

The original codes needs additional specification of hyperparameters to start, as documented in Appendix A.1. Specifically, we had to initialize the latent space configuration to utilize spherical sampling to match the Progressive GAN geometry, and manually enable memory cache clearing to prevent Out-Of-Memory (OOM) errors on the RTX 4090 hardware. Furthermore, we had to reconcile deprecated *Torchvision* arguments (specifically transitioning from `range` to `value_range`) to ensure the visual outputs were correctly normalized between [-1, 1]. These adjustments provided the baseline stability necessary to initiate the 48-hour training process.

Hardware Setup:

- **Compute:** 4x NVIDIA RTX 4090 GPUs (Distributed Data Parallel).
- **Training Duration:** ∼48 hours.

4.2 Technical Barriers: The Drift Penalty

During the replication of the VAE-GAN pipeline, we encountered a severe training instability characterized by immediate model collapse when the training progresses to the stage of resolution 64×64. Analysis of the latent space revealed that the variance ($Var(z)$) spiked to approximately 5.9×10^6. This "Exploding Variance" phenomenon occurred because the Encoder attempted to "cheat" the reconstruction objective: by pushing latent embeddings (z) toward infinity, it maximized the distance between different designs, making them easier for the Generator to distinguish and reconstruct. This behavior, however, led to numerical overflow (NaNs) and frequent CUDA assertion errors during the mask loss calculation.

To mitigate this, we introduced a **Drift Penalty** ($\mathcal{L}_{drift}$) as a regularization term to act as a "leash" on the Encoder:

$$\mathcal{L}_{drift} = \epsilon \cdot ||z||^2 \tag{7}$$

As detailed in Appendix A.1, the initial penalty was set at $\epsilon = 10^{-3}$, but failure analysis from training logs (specifically *log_131825*) indicated that a more precise constraint was required to prevent immediate saturation. We subsequently tuned the hyperparameter down to $\epsilon = 10^{-5}$ to balance stability with reconstruction accuracy.

While this regularization successfully stabilized the variance—dropping it from 10^7 to approximately 3×10^5—it revealed a fundamental "Stability vs. Generation" trade-off. The penalty was sufficient to stop model crashes and allow for successful reconstructions where $z \approx \pm 560$. However, it remained too weak to force the latent distribution into the Standard Normal space ($\mathcal{N}(0, 1)$) required for sampling. Consequently, although the model could "photocopy" existing designs, the magnitude mismatch between the "loud" training signals and the standard noise ($z \approx 1$) resulted in the generative failure observed in the final stages.

4.3 Replication Results: The "White Screen" Phenomenon

Despite the successful stabilization of the training process through the drift penalty, the model exhibited a fundamental failure in its generative capability, a state we identify as the "White Screen Phenomenon". This failure is characterized by a stark divergence between the model's ability to memorize existing data and its ability to synthesize new designs.

1. **Reconstruction (Memorization):** As shown in Figure 1, the model achieved successful reconstructions of the training set (e.g., `reconstructions_phase4_iter4376960.png`). In this mode, the Encoder effectively "photocopies" the input by generating massive latent signals. Evidence from Batch 3801 confirms that the latent variance exploded to approximately $314,654$, meaning the Generator learned to respond to signals in the magnitude of $z \approx \pm 560$.

2. **Random Sampling (Generative Failure):** When attempting to generate novel designs by sampling from standard normal noise ($z \sim \mathcal{N}(0, 1)$), the model failed completely. The resulting output was a field of pure white pixels, indicating a total lack of feature activation.

Root Cause Analysis: The technical bottleneck was a severe *signal magnitude mismatch*. While the Drift Penalty prevented a numerical crash, it was too weak to force the latent space into a Standard Normal distribution. Because the Generator was optimized to interpret "loud" signals ($z \approx 560$), the standard noise input ($z \approx 1$) represented only $\sim 0.17\%$ of the expected signal magnitude. Consequently, the input was insufficient to trigger the neural pathways required for image synthesis, leading to a default bias output. This result confirms that training complex VAE-GAN architectures from scratch on small datasets is highly brittle and often results in posterior collapse.

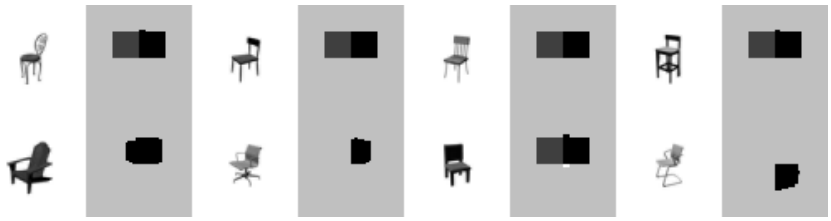

Figure 1: Reconstructions of chairs with random samples

5 Extension: A Foundation Model Agentic Workflow

In light of the convergence instabilities and computational overhead associated with the classical VAE-GAN framework, we propose an alternative paradigm: a **Foundation Model Agent System**. This approach shifts the technical focus from training custom generative architectures to orchestrating specialized, pre-trained agents that leverage multi-modal reasoning and zero-shot capabilities.

5.1 The Multi-Agent Architecture

We developed an integrated system comprising three specialized agents designed to simulate the iterative product development lifecycle:

- **Designer Agent (Stable Diffusion):** Utilizing a latent diffusion foundation model, this agent serves as the generative engine, producing high-fidelity, high-resolution design prototypes based on semantic prompts.
- **Market Agent (CLIP):** Acting as a zero-shot evaluator, this agent utilizes Contrastive Language-Image Pre-training (CLIP) to align visual designs with marketing-relevant semantic concepts.
- **Operations Agent (Computer Vision):** A rule-based diagnostic agent that employs computer vision algorithms to quantify manufacturing complexity, serving as a proxy for production feasibility and cost.

5.2 Market Agent: Zero-Shot Semantic Scoring

To overcome the "thin data" barrier of training a custom regressor, the Market Agent utilizes CLIP's shared embedding space to quantify aesthetic attributes. We implemented a **5-Point Semantic Scale** using five anchor prompts ranging from "Traditional Antique" (valued at 1.0) to "Futuristic Modern" (valued at 5.0).

The agent calculates a continuous aesthetic score as the **expected value** of the softmax probability distribution across these semantic buckets:

$$\text{Score} = \sum_{i=1}^{5} (P(\text{prompt}_i|\text{Image}) \times \text{ScaleValue}_i) \tag{8}$$

To mitigate errors stemming from 2D camera occlusions, we implemented a robust multi-view ensemble, averaging the scores derived from four distinct camera angles $(0°, 15°, 30°, 45°)$. This automated scoring mechanism was validated against human ground-truth ratings, yielding a significant positive correlation $(r = 0.3167, p < 0.001)$. This result suggests that the agent can successfully emulate consumer perception to perform high-volume design screening without human intervention.

5.3 Operations Agent: Structural Complexity as a Manufacturing Proxy

To reconcile aesthetic ambition with production feasibility, we developed a diagnostic Operations Agent designed to automate manufacturing cost estimation. The agent employs a computer vision-based proxy: structural edge density. By applying the Canny edge detection algorithm, the agent quantifies the frequency of contours and structural lines within a design. The underlying economic logic rests on the assumption that a high density of edges is a reliable indicator of manufacturing intensity.

$$\text{Edge Density} \propto \text{Manufacturing Difficulty} \approx \text{Sum of Cuts, Joints, \& Labor} \tag{9}$$

In practice, a minimalist design (e.g., a molded plywood chair) yields low edge density, whereas a highly intricate design (e.g., a complex woven rattan chair) produces thousands of detected edges. The agent normalizes these pixel-edge-sums into a relative Difficulty Index ranging from 0 to 100. This automated filtering allows the system to identify "concept-only" designs that, while aesthetically superior, would be prohibitively expensive to mass-produce due to excessive assembly labor or material complexity.

5.4 Results: Identifying the Pareto-Efficient Frontier

The integration of the Market and Operations agents enabled a large-scale, automated evaluation of 700 design candidates. By plotting the semantic modernity scores against the manufacturing complexity index, we identified a Pareto-efficient frontier—a subset of designs representing the optimal trade-off between market appeal and production feasibility. As illustrated in Figure 2, this frontier is defined by two rigorous strategic constraints:

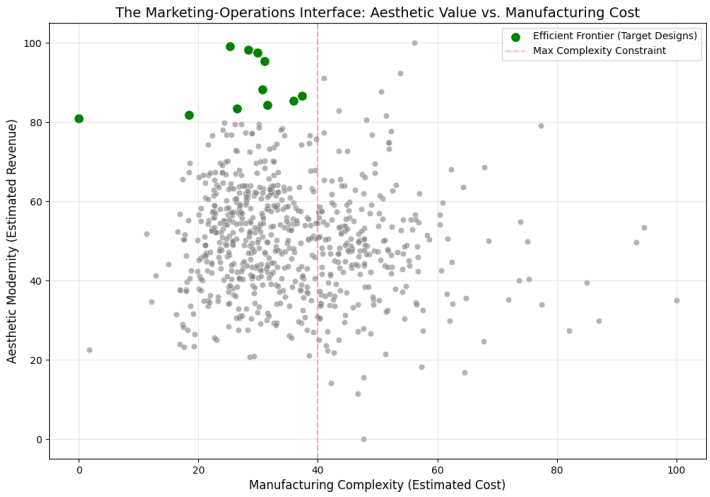

Figure 2: The Efficient Frontier: Plotting Modernity vs. Manufacturing Complexity for 700 designs. The green zone represents the strategic "Sweet Spot".

- **Marketing Constraint (High Modernity):** A Modernity Score exceeding 80, which filters for designs with high perceived revenue potential based on CLIP-based semantic alignment.
- **Operations Constraint (Low Complexity):** A Complexity Score below 40, ensuring that selected designs remain within manageable manufacturing cost parameters.

By deploying this dual-agent screening, the system successfully identified a "Sweet Spot": a portfolio of designs that maximize aesthetic modernization while simultaneously minimizing production complexity. This automated discovery process effectively reduced a vast, unstructured design space into a manageable selection of high-probability market successes without requiring expensive human-led clinics or early-stage engineering feasibility studies.

5.5 Visual Analysis of Automated Discovery

To validate the efficacy of the agentic workflow, we conducted a visual comparison of designs located at different points within the complexity-modernity space. This analysis contrasts catalog-existing items with a synthetic prototype generated by the Designer Agent.

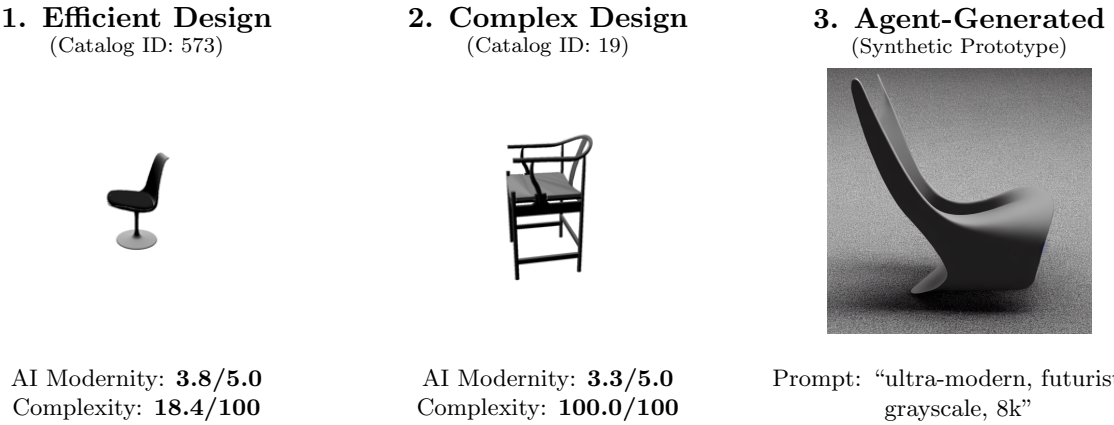

Figure 3: Comparative visual analysis of design candidates. The system distinguishes between efficient, low-cost designs (Left) and traditional but prohibitively complex alternatives (Center), while providing generative capabilities for new iterations (Right).

The "Efficient Design" (ID: 573) exemplifies the Pareto-optimal selection, achieving a high AI modernity rating of 3.8/5.0 while maintaining a low complexity score of 18.4. In contrast, the "Complex Design" (ID: 19) represents a significant operational risk. This chair is not only highly traditional rated by AI, but also with structural complexity index reaches the maximum value of 100.0, indicating excessive manufacturing difficulty. Finally, the "Agent-Generated" prototype demonstrates the utility of the Foundation Model Agent in synthesizing novel, high-modernity designs that can be immediately re-evaluated by the Market and Operations agents to ensure they remain on the efficient frontier.

6 Conclusion

This report acknowledges the seminal theoretical contributions of Burnap et al. (2023) in establishing a quantifiable framework for product aesthetics. By bridging the gap between unstructured visual Gestalt and structured marketing metrics, the original work provided a crucial "ruler" for design valuation. Our replication attempt successfully validated these core conceptual pillars, confirming the potential for deep learning to translate aesthetic appeal into a strategic business asset.

However, the replication also highlighted specific practical challenges associated with the implementation of custom VAE-GAN architectures. In our execution, the model exhibited significant training instability, most notably characterized by latent variance explosion and a subsequent "White Screen" generative failure. While these outcomes are specific to this replication and may be influenced by localized hyperparameter settings and hardware environments, they underscore the inherent complexity of training high-dimensional generative models from scratch.

To address these limitations, we proposed an Agentic Workflow utilizing foundation models such as Stable Diffusion and CLIP. By shifting the technical focus from custom pixel-level training to the orchestration of pre-trained agents, we demonstrated a more robust and scalable alternative for aesthetic research. The integration of the Market Agent for semantic scoring and the Operations Agent for complexity mapping provides a closed-loop system for automated product discovery. Ultimately, this augmented approach preserves the theoretical spirit of the original research while offering a computationally stable path toward identifying the Pareto-efficient "Sweet Spot" of product design.

7 References

Burnap, A., Hauser, J. R., & Timoshenko, A. (2023). Product aesthetic design: A machine learning augmentation. *Marketing Science*, *42*(6), 1029–1056. https://doi.org/10.1287/mksc.2022.1429

A Technical Implementation and Stability Fixes

A.1 Appendix A: Fixes for Replication Stability

During the replication of the VAE-GAN pipeline, several critical technical hurdles were addressed to ensure training stability and numerical accuracy.

1. Drift Penalty (Preventing Variance Explosion) We introduced a drift penalty to penalize the encoder if latent output magnitudes (z) grew too large. This was tuned specifically to address the failures observed in the original log analysis.

```
# In train.py (Encoder Step)
# We penalize the encoder if raw outputs (real_h) get too large.
# Tuned from 0.001 -> 0.00001 based on log_131825 failure analysis.

drift_loss = (real_h ** 2).mean() * 0.00001
e_losses.append(drift_loss)
```

Listing 1: Encoder Drift Penalty Implementation

2. Numerical Stability Fixes To prevent system crashes (specifically CUDA assertion errors) and handle multi-GPU gradient clipping, the following adjustments were made:

```
# A. Fix CUDA Assertion Error in Mask Loss
# Squash Tanh output (-1, 1) into a Sigmoid-like range (0, 1)
mask_loss = mask_loss_criterion(torch.sigmoid(recon_mask), batch_masks)

# B. Fix Gradient Clipping for Distributed Data Parallel (DDP)
# Direct parameter access bypasses generator graph detachment issues
if labeled_data:
    p_loss.backward()
    torch.nn.utils.clip_grad_norm_(encoder.parameters(), max_norm=1.0)
    torch.nn.utils.clip_grad_norm_(predictor.parameters(), max_norm=1.0)
```

Listing 2: Loss and Gradient Stability Fixes

3. Latent Space Sampling (Spherical Normalization) Standard ProGAN architectures often benefit from sampling from a hypersphere rather than a standard Gaussian cube. We enforced spherical normalization on the random noise z.

```
# In config.py: Enforce spherical latent space configuration
c.noise = 'sphere'

# In evaluate.py: Project random noise onto the sphere
myz = Variable(torch.randn(colN, session.c.nz)).cuda()
myz = training_utils.normalize(myz)
```

Listing 3: Spherical Sampling Logic

4. Environment Compatibility Settings were adjusted to prevent memory fragmentation and ensure compatibility with modern versions of the Torchvision library.

```
# In config.py: Prevent GPU Out-of-Memory (OOM) fragmentation
c.empty_cache = True

# In evaluate.py: Fix for newer torchvision versions (value_range)
torchvision.utils.save_image(
    torch.cat(images, 0),
    save_path,
    nrow=colN,
```

```
    normalize=True,
    value_range=(-1, 1),
    padding=0
)
```

Listing 4: Memory and Library Compatibility

B Agentic AI Logic and Scoring

B.1 Appendix B: Market Agent Logic (CLIP Scoring)

The Market Agent calculates an "Expected Value" for aesthetic modernity by evaluating the probability distribution across a defined 5-point semantic scale.

```
# 1. Define Semantic Anchors for 5-Point Scale
SCALE_MAPPING = {
    "a photo of a very traditional antique chair": 1.0,
    "a photo of a traditional chair": 2.0,
    "a photo of a standard chair": 3.0,
    "a photo of a modern chair": 4.0,
    "a photo of a futuristic modern chair": 5.0
}

# 2. Multi-View Ensemble (Robustness)
# Average probabilities across 4 angles (0, 15, 30, 45) for stability
view_probs = []
for v in VIEWS:
    logits = model(text=PROMPTS, images=img_view[v])
    probs = logits.softmax(dim=1)
    view_probs.append(probs)

# 3. Calculate Final AI Rating
avg_probs = np.mean(view_probs, axis=0)
ai_rating = np.sum(avg_probs * [1.0, 2.0, 3.0, 4.0, 5.0])
```

Listing 5: CLIP-based Weighted Semantic Scoring

B.2 Appendix C: Operations Agent Logic (Complexity)

The Operations Agent utilizes edge density as a structural proxy for manufacturing labor and assembly difficulty.

```
def calculate_complexity(img_array):
    # Detect structural lines (cuts/joints) using Canny Edge Detection
    edges = cv2.Canny(img_array, 100, 200)
    edge_score = np.sum(edges) / 255.0
    return edge_score

# Main Analysis Loop with Normalization
complexities = []
for img in dataset:
    edge_score = calculate_complexity(img)
    complexities.append(edge_score)

# Normalize raw scores to 0-100 scale
df['ops_score'] = (raw - min) / (max - min) * 100
```

Listing 6: Manufacturing Complexity Proxy